

MARVINO MARULI

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PROFESSIONAL SUMMARY

Third-year Computer Science student at Bina Nusantara University (BINUS) focused on backend engineering. Built transactional savings APIs in NestJS and PostgreSQL for a five-person personal-finance app. Co-authored container vs. serverless performance research presented at EECSI 2026 (IEEE-sponsored). Skilled in REST API design, database concurrency control (isolation levels, row locking, append-only ledgers), and reproducible benchmarking with statistical analysis. Seeking a Software Engineering internship building reliable backend systems.

CORE SKILLS

Backend Development | REST API Design | PostgreSQL | Concurrency Control | Unit Testing | Performance Benchmarking | Containerization | Technical Writing

EDUCATION

Bachelor of Computer Science

Expected 2028

Bina Nusantara University (BINUS), Jakarta, Indonesia | GPA: 3.53 / 4.00

Academic project: Software Architecture final project for Harmony. Documented creational, structural and behavioral design patterns; compared layered, microservice and pipeline architectures; evaluated the system against ISO/IEC 25010.

PROFESSIONAL EXPERIENCE

BACKEND DEVELOPER

2026

Harmony: Personal Finance App (5-person university team project, BINUS)

- Designed the backend API specification across **7 modules** for a NestJS 11, TypeScript and PostgreSQL stack, with a Flutter client and a Python/Flask OCR microservice.
- Developed the savings module (deposits, withdrawals, savings-goal CRUD, wallet-integrated savings) using **SERIALIZABLE transactions and pessimistic row locks** (SELECT ... FOR UPDATE) to prevent concurrent overdrafts, with append-only progress and withdrawal logs for auditability; covered by **~50 unit tests**.
- Fixed an OCR receipt-parsing bug that misread quantities (e.g., "1 500 ml x 7,000" parsed as qty 500), reshaped the OCR API contract, and ported the service to Hugging Face Spaces.

UNDERGRADUATE RESEARCHER & CO-AUTHOR

2026

"Docker vs. Serverless" Performance Benchmark, presented at EECSI 2026 (IEEE-sponsored), Yogyakarta

- Built a reproducible benchmark of Docker containers vs. locally emulated serverless execution across multiple workloads; open-sourced at github.com/vinnsssss3/Docker-vs-Serverless.
- Measured **0.01–1.30 ms** median warm-state latency gaps, **64–82 ms** cold-start overhead, and **~57%** lower cumulative memory-time cost for serverless under sparse traffic.
- Applied TOST equivalence testing, bootstrap confidence intervals and p99 latency analysis; wrote the Results and Discussion chapters and addressed **5 rounds** of reviewer feedback.

SELECTED PROJECTS

PERSONAL PORTFOLIO WEBSITE

Sept 2026

vinnsssss3.github.io | HTML, CSS, JavaScript | Deployed on GitHub Pages

- Built a framework-free, no-build-step site; content lives in data files separate from rendering code.
- Implemented light and dark themes, reduced-motion support, WCAG AA text contrast, deep-linkable case-study dialogs, and a canvas-rendered interactive ASCII animation.
- Verified at 1440 px and 390 px widths in both themes: zero console errors, no horizontal overflow.

AWARDS AND ACHIEVEMENTS

- Conference presenter, EECSI 2026 (IEEE-sponsored), Yogyakarta, 2–3 September 2026
- Open-source research artifact: public, reproducible benchmark repository accompanying the paper

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Dart

Backend & Databases: NestJS, Node.js, Express, Flask, TypeORM, Prisma | PostgreSQL, MySQL

DevOps, Cloud & Testing: Docker, Git/GitHub, GitHub Pages, Vercel, Hugging Face Spaces | Jest (unit testing)

Other: Flutter (familiar), HTML/CSS

LANGUAGES

Indonesian (Native) | English (Professional working proficiency)

ADDITIONAL INFORMATION

Professional interests: Backend systems, database transaction safety, system performance, distributed systems

REFERENCES

Available upon request.